

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

Nov 2012

CL201 Surveying

Date: 27.11.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain principal of plane table and list the tools and equipment used in plane table survey. [04]
- (b) Differentiate between EDM and total-station. [03]
- Q - 2 (a) List the methods of plane tabling, explain radiation method in detail [04]
- (b) Explain procedure to measure included angle with the help of theodolite. [04]
- (c) Calculate and correct the latitudes and departures for the following data [06]

Line	Length	RB
AB	255	S 39° 18' E
BC	656	N 34° 58' 55'' E
CD	120	N 21° 18' 34'' E
DA	668	S 47° 23' 13'' W

OR

- Q - 2 (a) List the errors in plane table survey. [04]
- (b) Explain temporary adjustment of theodolite. [04]
- (c) Explain all rules of balancing of traverse. [06]
- Q - 3 (a) Write advantages and disadvantages of plane table survey. [04]
- (b) Write the function of the following [04]
- i) Foot Screw ii) Tangent screw iii) Graphic display iv) ATR
- (c) What is setting out of building? Explain procedure to set out building with neat sketch. [06]

OR

- Q - 3 (a) Explain with sketch method of intersection, also mention when this method is suitable. [04]
- (b) Define following [04]
- i) Telescope normal ii) Line of collimation iii) Total station iv) EDM

- (c) Explain procedure to set out culvert with neat sketch. [06]

SECTION – II

- Q - 4 (a) Write sources of errors in leveling [04]

- (b) Draw the following contour [03]

i) Hill ii) Vertical cliff iii) Saddle

- Q - 5 (a) i) Draw the sketch showing elements of simple circular curve. [08]

ii) List the methods to set out simple circular curve with help of chain and tape only. Explain any one in detail.

- (b) Calculate area by trapezoidal rule and Simpson's rule for the following data [06]

Chainage(m)	0	15	30	45	60	75	90	105	120	135
Offset (m)	3.5	4.30	6.75	5.25	7.50	8.80	7.90	6.40	4.40	3.25

OR

- Q - 5 (a) Explain field procedure of setting out of simple circular curve by one theodolite. [08]

- (b) The area of a square of 10cm side was measured by a planimeter with the anchor point outside the figure and the initial and final readings were found to be 6.852 and 8.704, respectively. With the same setting of the tracing arm and the anchor point outside, another irregular figure was traversed clockwise and the initial and final readings were found to be 2.378 and 8.626, respectively. What is the area of the figure? If plain scale is 1cm = 10m, Calculate area in m^2 . [06]

- Q - 6 (a) What is sounding? Explain procedure of echo sounder with neat sketch. [04]

- (b) An embankment of width 10m and side slopes 1.5:1 is required to be made on a ground which is level in a direction transverse to the centre line. The central height at 40m intervals are as follows. [06]

0.90, 1.25, 2.15, 2.50, 1.85, 1.35, and 0.85

Calculate the volume of earth work according to (i) the trapezoidal formula and (ii) the prismoidal formula.

- (c) If a curve is designated as a 4° curve on a 30 m. arc, find the tangent distance, length of long chord, length of arc, apex distance and mid ordinate if the deflection angle is 36° [04]

OR

- Q - 6 (a) Explain sounding procedure with help of sounding cable and sounding rod. [04]

- (b) A road at the formation level is 6m wide and has a side slope of 2:1. The road is to have a constant R.L of 200m. The ground is level across the centre line of the road. The following observations were made. Estimate the volume of earth work [06]

Chainage(m)	0	20	40	60	80	100
Surface levels along c/l road	204.6	203	200.8	201.6	202	200.2

- (c) If the tangent distance of curve is 235.6 m and the deflection angle 42° , calculate the [04]
other parameters of the curve, such as length of long chord, length of curve, apex
distance and mid ordinate.
